CCIT Report #532 May 2005

Relaying Protocols for Two Co-located Users

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Abstract

We consider a wireless network where a remote source wishes to send information to one of two co-located users, and where the second user can serve as a relay. The source's transmission is subjected to quasi-static flat Rayleigh fading, while the transmissions of the users experience a fixed amplitude gain with uniform random phase, capturing their mutual proximity. All communications share the same time/bandwith resources, and channel state information is known only to the receivers. We propose relaying protocols which are based on Wyner-Ziv quantization at the relay, and demonstrate their high efficiency (in terms of expected throughput) with respect to previously reported relaying schemes based on *amplify-and-forward* and *decode-and-forward*. We further incorporate into the cooperative scheme the notion of *successive quantization* with respect to the relay's observation, and match it with a broadcast transmission approach in the relay to destination link, relying on the inverse relation which exists between the amount of side information at the destination, and the capacity of the relay to the destination link. Finally, we consider a hybrid *amplify-quantize-decode-and-forward* scheme which exhibits superior performance in some situations.

Keywords— Ad-hoc networks, amplify and forward, broadcast strategy, cooperative diversity, decode and forward, expected throughput, fading channels, outage capacity, quantize and forward, relay channel, sensor networks, successive refinement, wireless networks.

I. Introduction

Cooperative strategies for wireless networks are drawing a lot of attention in recent years, mainly due to the proliferation of stand alone wireless networks which do not rely on any fixed infrastructure to facilitate the communications between the nodes in the network. Moreover, these networks are typically employed in a fading environment, rendering the transmitted signals susceptible to severe attenuations in

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The work was supported by the Fund of Promotion of Research at the Technion and the ISRC consortium.